

18-19

力与运动

1. 圆周运动角量描述

$$\theta = \int \omega dt = (\omega - \omega_0) \cdot \beta = \omega_0 t + \frac{1}{2} \beta t^2$$

$$\omega = \int \beta dt = \frac{d\theta}{dt} = \beta t + \omega_0$$

$$\beta = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$a = \frac{d\vec{v}}{dt} = \frac{d(v\vec{e}_r)}{dt} = \frac{dv}{dt}\vec{e}_r + v\frac{d\vec{e}_r}{dt} = \frac{dv}{dt}\vec{e}_r + v\frac{d\theta}{dt}\vec{e}_\theta = \frac{dv}{dt}\vec{e}_r + \frac{v^2}{r}\vec{e}_\theta$$

$$a = \sqrt{a_r^2 + a_\theta^2} \quad \theta = \arctan \frac{a_\theta}{a_r}$$



$$\frac{d\theta}{dt} = \frac{d\theta}{dl} \frac{dl}{dt} = \frac{v}{r}$$

$$a_r = \frac{dv}{dt} = \beta r = \beta R$$

$$a_\theta = \frac{v^2}{r} = \frac{v^2}{R} = \omega^2 R$$

$$\begin{aligned} \vec{v} &= \vec{\omega} \times \vec{r} \\ \vec{a}_n &= \vec{\omega} \times (\vec{\omega} \times \vec{r}) \\ &= -\vec{\omega} \times \vec{v} \\ \vec{a}_t &= \vec{\beta} \times \vec{r} \end{aligned}$$

2. 直线运动 + 变力做功 → 动能定理

$$x = \int v dt = \frac{1}{2a}(v^2 - v_0^2) = v_0 t + \frac{1}{2}at^2$$

$$v = \frac{dx}{dt} = \int a dt = v_0 + at$$

$$a = \frac{dv}{dt} = \frac{dx}{dt} \frac{dv}{dx} = v \frac{dv}{dx}$$

$$F(t) = m \frac{dv}{dt} \Rightarrow \int_{v_0}^{v_t} dv = \int_{t_0}^{t_t} \frac{1}{m} F(t) dt$$

$$F(v) = m \frac{dv}{dt} \Rightarrow \int_{v_0}^{v_t} \frac{F(v)}{v} dv = \int_{t_0}^{t_t} \frac{1}{m} dt$$

$$F(x) = m \frac{dv}{dt} \Rightarrow \int_{x_0}^{x_t} \frac{F(x)}{m} dx = \int_{v_0}^{v_t} v dv$$

$$\Delta E_k = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \int F dx = A$$

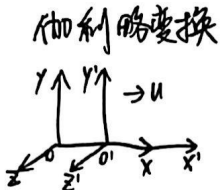
3. 冲量与动量

$$I = \int F dt \quad p = mv \quad \Delta p = p - p_0 = I$$

$$F = \frac{dI}{dt} \quad \vec{F} = \frac{1}{t-t_0} \int_{t_0}^t \vec{F}(t) dt \quad \sum \vec{F}_{\text{外}} = 0 \Rightarrow p \text{ 守恒}$$

4. 相对论

△ 绝对时空观



$$\begin{cases} x = x - ut \\ y = y' \\ z = z' \\ t = t' \end{cases} \quad \begin{cases} u_x' = x - u \\ u_y' = u_y \\ u_z' = u_z \end{cases}$$

麦克斯韦 → “以太” → 迈克尔逊-莫雷

△ 狭义相对论基本原理

○ 洛伦兹变换

$$\begin{aligned} \text{正: } \begin{cases} x' = (x - ut)k \\ x = (x' + ut')k \\ x = ct \\ x' = ct' \end{cases} & \quad \begin{cases} ct' = [c^2t^2 - ut(x-x') - u^2t'^2]^{1/2} \\ ct = [c^2t'^2 - uct'(t'-t) + u^2t^2]^{1/2} \\ ct' = [c^2t^2 - uctt' + uct'^2 - u^2t'^2]^{1/2} \\ ct = [c^2t'^2 - uct't + uct^2 - u^2t^2]^{1/2} \end{cases} \\ \Rightarrow k = \frac{1}{\sqrt{1-u^2/c^2}} & \quad = k^2 t' (c^2 - u^2) \end{aligned}$$

$$\begin{cases} x' = \frac{x - ut}{\sqrt{1-u^2/c^2}} \\ y' = y \\ z' = z \\ t' = \frac{t - ux/c^2}{\sqrt{1-u^2/c^2}} \end{cases} \quad \text{逆} \quad \begin{cases} x = \frac{x' + ut'}{\sqrt{1-u^2/c^2}} \\ y = y' \\ z = z' \\ t = \frac{t' + ux'/c^2}{\sqrt{1-u^2/c^2}} \end{cases}$$

○ 爱因斯坦速度变换

$$\begin{cases} u_x' = \frac{u_x - u}{1 - u u_x / c^2} \\ u_y' = \frac{u_y \sqrt{1-u^2/c^2}}{1 - u u_x / c^2} \\ u_z' = \frac{u_z \sqrt{1-u^2/c^2}}{1 - u u_x / c^2} \end{cases} \quad \text{逆} \quad \begin{cases} u_x = \frac{u_x' + u}{1 + u u_x' / c^2} \\ u_y = \frac{u_y' \sqrt{1-u^2/c^2}}{1 + u u_x' / c^2} \\ u_z = \frac{u_z' \sqrt{1-u^2/c^2}}{1 + u u_x' / c^2} \end{cases}$$

△ 狭义相对论时空观 K

$$\text{① “同时”相对: } \begin{matrix} A & t_1 = t \\ B & t_2 = t \end{matrix} \quad \begin{matrix} K' & u \\ t_1' & = \frac{t - ux_1/c^2}{\sqrt{1-u^2/c^2}} \\ t_2' & = \frac{t - ux_2/c^2}{\sqrt{1-u^2/c^2}} \end{matrix}$$

$$\text{② 长度收缩: 同时 } \begin{matrix} K \text{ 或 } K' \text{ 中 } & \text{仅当 } x_1 = x_2 \text{ 时 } t_1' = t_2' \\ l_0 = x_1 - x_2 & x_1' = \frac{x_1 - ut_1'}{\sqrt{1-u^2/c^2}} \quad x_2' = \frac{x_2 - ut_2'}{\sqrt{1-u^2/c^2}} \\ \text{同时测} & l' = (x_1 - x_2) \sqrt{1-u^2/c^2} \\ \text{相对静止的系中为 } l_0 \text{ max} & \end{matrix}$$

$$\text{③ 时间膨胀: 同地 } \Delta t_0 = t_2 - t_1 \quad \Delta t' = \frac{t_2 - ux_2/c^2}{\sqrt{1-u^2/c^2}} - \frac{t_1 - ux_1/c^2}{\sqrt{1-u^2/c^2}} = (t_2 - t_1) / \sqrt{1-u^2/c^2}$$

$$\text{④ 两事件不同时不同地 - 洛伦兹收缩 } \Delta x < \Delta t$$

⑤ 时空间隔绝对

$$S = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2(t_2 - t_1)^2}$$

因果时序绝对

$$t_2 - t_1 = \frac{(t_2' - t_1')}{\sqrt{1-u^2/c^2}} \left(1 + \frac{u}{c^2} \frac{x_2' - x_1'}{t_2' - t_1'} \right) > 0$$

$$\begin{aligned} x_2 - x_1 &= \frac{(x_2' - x_1') + ut(t_2' - t_1')}{\sqrt{1-u^2/c^2}} = \frac{\Delta x' + u \Delta t'}{\sqrt{1-u^2/c^2}} \\ t_2 - t_1 &= \frac{(t_2' - t_1') + u(x_2' - x_1')/c^2}{\sqrt{1-u^2/c^2}} = \frac{\Delta t' + u \Delta x'/c^2}{\sqrt{1-u^2/c^2}} \end{aligned}$$



5. 角动量 + 刚体定轴转动

$$\vec{M} = \vec{r} \times \vec{F}$$

$$\vec{M} = J \vec{\beta} \leftarrow \sum (F_{\text{切}} + F_{\text{法}}) r_i = \sum m_i \beta r_i r_i$$

$$J = \sum m_i r_i^2 = \int r^2 dm$$

质点系

$$J = J_0 + m h^2$$

$$J_z = J_x + J_y$$

$$R_G = \sqrt{\frac{J}{m}} \quad J = m R_G^2$$

$$E_k = \sum \frac{1}{2} m_i v_i^2 = \frac{1}{2} \sum m_i r_i^2 \cdot \omega^2$$

$$= \frac{1}{2} J \omega^2$$

$$m = J \frac{d\omega}{dt} = J \frac{d\omega}{d\theta} \frac{d\theta}{dt} = J \omega \frac{d\omega}{d\theta}$$

$$A = \int F dr = \int M d\theta = \int J \omega d\omega = \frac{1}{2} J \omega^2 - \frac{1}{2} J \omega_0^2$$

$$M = \frac{d(J\omega)}{dt} \Rightarrow \int_0^t M dt = J\omega - J\omega_0 \quad \sum \vec{M} = 0 \Rightarrow J\omega \text{ 守恒}$$

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times (m\vec{v})$$

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} = \vec{v} \times \vec{p} + \vec{r} \times \vec{F}$$

$$\Rightarrow \vec{M} = \frac{d\vec{L}}{dt}$$

$$\int_{t_0}^t \vec{M} dt = \vec{L} - \vec{L}_0 \quad \sum \vec{M} = 0 \Rightarrow \vec{L} \text{ 守恒}$$

对比

平动

$$m \times v \propto F$$

$$\vec{F} = m\vec{a}$$

$$\vec{p} = m\vec{v}$$

$$E_k = \frac{1}{2} m v^2$$

$$\int \vec{F} dt = \Delta \vec{p}$$

转动

$J \theta \propto M L$

$$\vec{M} = J \vec{\beta} = \vec{r} \times \vec{F}$$

$$\vec{L} = J \vec{\omega} = \vec{r} \times \vec{p} = \vec{r} \times (m\vec{v})$$

$$E_k = \frac{1}{2} J \omega^2$$

$$\int \vec{M} dt = \Delta \vec{L}$$

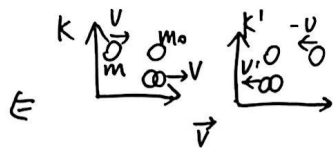
6. 相对论动力学

$$\text{质速 } m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt} \left(\frac{m_0 \vec{v}}{\sqrt{1 - v^2/c^2}} \right)$$

$$\text{质动 } \vec{p} = m\vec{v} = \frac{m_0 \vec{v}}{\sqrt{1 - v^2/c^2}}$$

$$\boxed{\text{光子}} \quad E = h\nu = hc/\lambda \quad m = \frac{E}{c^2} = \frac{h\nu}{c^2} \quad p = \frac{E}{c} = \frac{h}{\lambda}$$



$$M = m + m_0 \quad M' = m + m_0$$

$$m v = M V \quad m' (-v') = m (-v)$$

$$-v' = \frac{v - v}{1 - v^2/c^2}$$

质能

$$dE_k = \vec{F} \cdot d\vec{r} = \frac{d(m\vec{v})}{dt} \cdot \vec{v} dt = c^2 dm$$

$$E_k = mc^2 - m_0 c^2 = m_0 c^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)$$

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} \quad (v/c) = \left(\frac{pc}{E} \right)^2$$

$$\text{能动 } E^2 = \frac{m_0^2 c^4}{1 - p^2 c^2 / E^2}$$

$$\Rightarrow E^2 = p^2 c^2 + m_0^2 c^4$$

7. 振动合成

$$\text{描述 } x = A \cos(\omega t + \varphi)$$

$$v = -A\omega \sin(\omega t + \varphi)$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\text{动力 } F = -kx = m \frac{d^2 x}{dt^2} \Rightarrow \frac{d^2 x}{dt^2} = -\frac{k}{m} x \text{ 简谐运动}$$

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} \quad \varphi = \arctan \frac{v_0}{\omega x_0}$$

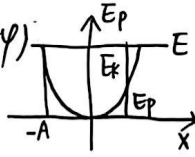
$$\text{能 } E_p = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi)$$

$$E_k = \frac{1}{2} m v^2 = \frac{1}{2} k A^2 \sin^2(\omega t + \varphi)$$

$$E = E_p + E_k = \frac{1}{2} k A^2$$

$$\frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$v = \pm \sqrt{\frac{k}{m} (A^2 - x^2)} \text{ 解1次}$$



$$\boxed{\text{扭摆}} \quad \tau = -k\theta = J \frac{d^2 \theta}{dt^2} \Rightarrow \frac{d^2 \theta}{dt^2} = -\frac{k}{J} \theta$$

(不同 φ)

$$\text{① 同向同 } x = A \cos(\omega t + \varphi)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\varphi = \arctan \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

② 同向不同 f (同 φ)

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos(\omega_2 - \omega_1)t}$$

$$\text{当 } A_1 = A_2 \quad A = 2A_1 \cos \frac{\omega_1 - \omega_2}{2} t \text{ 被调制 - 拍}$$

$$x = 2A_1 \cos \left(\frac{\omega_1 - \omega_2}{2} t \right) \cos \left(\frac{\omega_1 + \omega_2}{2} t \right)$$

$$\boxed{\text{拍频}} \quad \nu = \frac{\omega_1 - \omega_2}{2\pi} \times 2 = \frac{\omega_1 - \omega_2}{2\pi} = \nu_1 - \nu_2$$

③ 相互垂直

$$\varphi_1 = \varphi_2 \pm k\pi \quad x = \pm \frac{A_1}{A_2} x \quad r = \sqrt{A_1^2 + A_2^2} \cos \omega t$$

$$\Delta \varphi = \frac{\pi}{2} (2k+1) \quad x = A_1 \cos \omega t \quad \left(\frac{x}{A_1} \right)^2 + \left(\frac{y}{A_2} \right)^2 = 1$$

$$y = A_2 \sin \omega t$$

$\Delta \varphi$ 任意



18-19

振动与波

附近小振动

8. 波动微分方程

$$\omega = 2\pi f \quad f = \frac{1}{T} \quad \lambda = \frac{v}{f}$$

平面简谐波 $x \ v \ a \ T = \frac{2\pi}{\omega} = \frac{1}{f} \quad f = \frac{\omega}{2\pi} \quad \lambda = vT \quad k = \frac{2\pi}{\lambda}$ 波速 $u = \frac{\omega}{k}$ 相速

各种摆

$$\begin{aligned} \text{波函数 } y(x,t) &= A \cos(\omega t - \frac{2\pi}{\lambda} x) = A \cos(\omega t - kx) \\ &= A \cos 2\pi(\nu - \frac{x}{\lambda}) = A \cos 2\pi(\frac{t}{T} - \frac{x}{\lambda}) \\ &= A \cos \omega(t - \frac{x}{v}) \end{aligned}$$

x 正向
超前 落后稳定平衡处 $\frac{dE_p}{dx} = 0 \quad \frac{d^2E_p}{dx^2} > 0$ 单摆 $E_p = mgl(1 - \cos\theta)$

$$F_t = -\frac{dE_p}{dx} = -mg \sin\theta = -\frac{mg}{l}\theta$$

$$\Rightarrow \omega = \sqrt{\frac{g}{l}} \quad T = 2\pi\sqrt{\frac{l}{g}}$$

复摆 $T = -\frac{mgl \sin\theta}{J} = -J \frac{d^2\theta}{dt^2}$

$$\Rightarrow \omega = \sqrt{\frac{mgl}{J}} \quad T = 2\pi\sqrt{\frac{J}{mgl}}$$

$$\begin{aligned} \text{一维波动微分方程 } \frac{\partial^2 y}{\partial x^2} &= -(\frac{\omega}{v})^2 A \cos \omega(t - \frac{x}{v}) \\ \frac{\partial^2 y}{\partial t^2} &= -\omega^2 A \cos \omega(t - \frac{x}{v}) \Rightarrow \boxed{\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}} \end{aligned}$$

$$\text{绳上横波 } \Sigma F_y = F(\sin\theta_2 - \sin\theta_1) = F\left[\left(\frac{\partial y}{\partial x}\right)_2 - \left(\frac{\partial y}{\partial x}\right)_1\right] = F \frac{\partial^2 y}{\partial x^2} \Delta x = (\mu \frac{\partial^2 y}{\partial t^2}) \Delta x$$

$$\Rightarrow \frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F} \frac{\partial^2 y}{\partial t^2} \quad v = \sqrt{\frac{F}{\mu}}$$

$$\text{棒上纵波 } v = \sqrt{\frac{Y}{\rho}} \quad \text{杨氏模量 } Y = \frac{(\frac{F}{S})}{(\frac{\Delta l}{l_0})}$$

弹性
惯性
应力
应变

波的能量

$$\text{能量密度 } \Delta E_k = \frac{1}{2} \mu \omega^2 \left(\frac{\partial y}{\partial t}\right)^2 \quad \delta = \sqrt{\Delta x^2 + \Delta y^2} \approx \Delta x \approx \frac{1}{2} \left(\frac{\partial y}{\partial x}\right)^2 \Delta x$$

$$\Delta E_p = F \delta l = F \frac{1}{2} \left(\frac{\partial y}{\partial x}\right)^2 \Delta x \quad F = \mu v^2$$

$$\Rightarrow \Delta E = \Delta E_k + \Delta E_p = \mu A^2 \omega^2 \sin^2 \omega(t - \frac{x}{v}) \Delta x$$

$$\text{线 } \left(\frac{\Delta E}{\Delta x}\right) = \mu A^2 \omega^2 \overline{\sin^2(t - \frac{x}{v})} = \frac{1}{2} \mu A^2 \omega^2$$

$$\text{体 } \bar{w} = \frac{1}{2} \rho A^2 \omega^2$$

$$\Delta t = \frac{\Delta x}{v} \quad \text{能流密度 } P = \frac{\Delta E}{\Delta t} = \mu A^2 \omega^2 \left[\sin \omega(t - \frac{x}{v})\right]^2$$

$$\bar{P} = \left(\frac{\Delta E}{\Delta x}\right) v$$

$$\text{波的强度 } I = \bar{w} v = \frac{1}{2} \rho A^2 \omega^2 v \quad E = \bar{I} S \Delta t$$

$$\text{球面波 } I = \frac{P_0}{4\pi r^2} \quad \frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \quad \frac{A_1}{A_2} = \frac{r_2}{r_1}$$

$$y(r,t) = \frac{A_0}{r} \cos \omega(t - \frac{r}{v})$$

波的叠加与干涉

$$\text{传播的独立性/叠加原理 } \frac{\partial^2 (y_1 + y_2)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 (y_1 + y_2)}{\partial t^2}$$

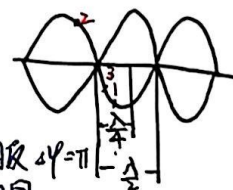
$$\text{相干波 (同/异/振动方向同)} \quad A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\varphi} \quad \Delta\varphi = (\varphi_2 - \varphi_1) - \frac{2\pi}{\lambda}(r_2 - r_1) = \begin{cases} \pm 2k\pi & \text{干涉相长} \\ \pm (2k+1)\pi & \text{干涉相消} \end{cases}$$

驻波 (A 同/相向)

$$\text{表达式 } y = A \left[\cos(\omega t + \frac{2\pi}{\lambda} x) + \cos(\omega t - \frac{2\pi}{\lambda} x) \right] = 2A \cos \frac{2\pi}{\lambda} x \cos \omega t$$

无振动状态/相位传播

无能量传播 = 总能量为 0, 相邻波节间动/势相互转换 分驻振动



$$\text{① 固定端 } x=L \text{ 处半波损失 } y = 2A \sin \frac{2\pi}{\lambda} x \cos \omega t$$

$$E = \frac{1}{2} \mu A^2 \omega^2 \lambda$$

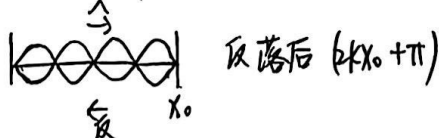
$$2\pi \frac{L}{\lambda} = n\pi \Rightarrow \lambda_n = \frac{2L}{n} \quad \nu_n = n \frac{v}{2L}$$

$$\text{基频 } \lambda_1 = 2L \quad \nu_1 = \frac{v}{2L} \quad \text{二次谐波 } \lambda_2 = L \quad \nu_2 = \frac{v}{L}$$

$$\text{② 自由端 } L = n \frac{\lambda}{4} \quad (n=1,3,5,\dots) \quad \nu_n = n \frac{v}{4L}$$

有 n 个波

$$\text{共振 (调节 } F/L/\mu) \quad \nu = \frac{v}{\lambda} = \frac{1}{2L} \sqrt{\frac{F}{\mu}}$$



9. 多普勒效应

接收到 $\nu_R = \frac{u+V_R}{u-V_S} \nu_S$ 2个V相同+远离-

电磁波无需介质 $\nu_R = \sqrt{\frac{c+u}{c-u}} \nu_S$

应用 红移/雷达/超声波测血流速

例题 ① 鸣笛而过一远离

② 山崖反射波长 $\lambda' = (u-v)T_S$
司机听到回声 $\nu_R = \frac{u+v}{u-v} \nu_S$

被反射回来的波与原发射波形成的拍频 $\nu_{拍} = \nu_S - \nu_R$

③ 媒质运动



$$\nu_R = \frac{\nu_{波入}}{\lambda'} = \frac{u+V_R}{u+V_R-V_S} \nu_S$$

$$\lambda' = (u+V_{波入})T_S$$

推导

$$\lambda = u/\nu_S$$

动
观察者

$$\nu_R = \frac{u+V_R}{\lambda} = \frac{u+V_R}{u} \nu_S$$

波源的 ν 不变

波源

$$\lambda' = \lambda - V_S T_S = (u - V_S) T_S$$

$$\nu' = \frac{u}{\lambda'} = \frac{u}{u - V_S} \nu_S \quad U_{波源}$$

$$\nu_R = \frac{u}{u - V_S} \nu_S$$

波源 ν 不变 媒质中 u 不变

△ 阻尼振动

$$x = A_0 e^{-\beta t} \cos(\omega' t + \varphi)$$

$$v = -A_0 \beta e^{-\beta t} \cos(\omega' t + \varphi) - A_0 \omega' e^{-\beta t} \sin(\omega' t + \varphi)$$

$$\beta = \frac{b}{2m} \quad \omega' = \sqrt{\omega_0^2 - \beta^2} \quad A_0 = \sqrt{x_0^2 + \left(\frac{\beta x_0}{\omega'}\right)^2} \quad \varphi = \arccos\left(\frac{x_0}{A_0}\right)$$

渐近性
时间常数 $\tau = \frac{1}{\beta} = \frac{m}{b}$
阻尼系数 β 减小为 $\frac{1}{2}$



△ 受迫振动

$$m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt} + F_0 \cos \omega t$$

稳态解 $x = A \cos(\omega t + \varphi)$
瞬态解

静电场

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2 \omega^2}} \quad \varphi = \arctan \frac{2\beta \omega}{\omega_0^2 - \omega^2}$$

$$\omega_{共振} = \sqrt{\omega_0^2 - 2\beta^2} \quad m \uparrow \quad \omega \downarrow \quad (应用)$$

12. 电场强度

电荷元叠加/高斯定理(对称)

$$\vec{E} = \int \frac{dq}{4\pi\epsilon_0 r^2} \vec{r}_i$$

$$\left\langle \begin{array}{l} \text{球} \quad \phi_e = 4\pi r^2 E \\ \text{柱} \quad \phi_0 = 2\pi r l \cdot E \\ \text{面} \quad \phi_0 = 20 S \cdot E \end{array} \right\rangle = \int_S \vec{E} \cdot d\vec{S} = \frac{\sum Q_{in}}{\epsilon_0}$$

模型

球体



$$E = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3} & r < R \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} & r > R \end{cases}$$

球壳

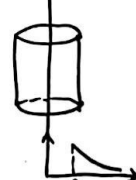
$$E = \begin{cases} 0 & r < R \\ \frac{Q}{4\pi\epsilon_0 r^2} & r > R \end{cases}$$

半球壳

$$dE = \frac{1}{4\pi\epsilon_0 R^2} (0 \cdot 2\pi R^2 \sin\theta d\theta) \cos\theta$$

$$E = \frac{\rho}{2\epsilon_0} \int_0^{\pi/2} \sin\theta \cos\theta d\theta = \frac{\rho}{4\epsilon_0}$$

圆筒



距轴 r 处

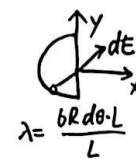
$$E = \begin{cases} 0 & r < R \\ \frac{\lambda l}{\epsilon_0 / 2\pi r l} = \frac{\lambda}{2\pi\epsilon_0 r} & r > R \end{cases}$$

半圆筒

轴线上

$$E_y = 0$$

$$E_x = \int \frac{1}{2\pi\epsilon_0} \frac{\lambda}{R} = \frac{\rho}{\pi\epsilon_0}$$



$$\lambda = \frac{\rho R d\theta \cdot L}{L}$$

圆柱

$$E = \left\{ \right.$$

无限长直导线

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad E = \frac{\lambda}{4\pi\epsilon_0 r}$$

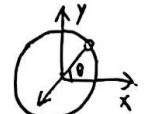
无限大平面

$$E = \frac{\rho}{2\epsilon_0}$$

$$\left\{ \begin{array}{l} \text{外} E=0 \\ \text{内} E=\frac{\rho}{\epsilon_0} \end{array} \right.$$

圆环

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda R d\theta}{R^2}$$



$$E_x = -\int_0^{2\pi} \frac{\lambda_0}{4\pi\epsilon_0 R} \cos\theta d\theta = -\frac{\lambda_0}{4\pi\epsilon_0 R}$$

$$E_y = -\int_0^{2\pi} \frac{\lambda_0}{8\pi\epsilon_0 R} \sin\theta d\theta = 0$$



23-24

热学

$$A = \int_{V_0}^{V_1} p dV = \frac{\Delta(PV)}{1-n}$$

多方 $pV^n = C$

17. 理想气体的多方过程 + 循环过程

	等p	等V	等T	绝热
$pV^n = C$				
多方指数	0	∞	1	γ
过程方程	$\frac{V}{T} = C$	$\frac{p}{T} = C$	$pV = C$	$pV^\gamma = C / V^{\gamma-1} T = C /$ $p^{1/\gamma} T = C$ $- \gamma (\frac{p}{T})$ 大
pV斜率	$-\alpha(\frac{p}{T})$	$-(1+\alpha)(\frac{p}{T})$	$-(\frac{p}{T})$	0
摩尔热容	$C_p = C_v + R$ 迈耶	$C_v = \frac{1}{2} R$	∞	0
气体对外做功(A)	$p \Delta V = \nu R \Delta T$	0	$\int_{V_0}^{V_1} p dV$	$- \nu C_v \Delta T = \frac{\Delta(PV)}{\gamma-1} = \frac{1}{\gamma-1} \Delta(PV)$
气体吸热Q(被)	$\nu C_p \Delta T$	$\nu C_v \Delta T$	$\nu R T \ln \frac{V_1}{V_0} = \nu R T \ln \frac{p_0}{p_1}$	0
气体内能增量 ΔE	$\nu C_v \Delta T$	$\nu C_v \Delta T$	0	$\nu C_v \Delta T$
$\Delta E = Q + A$	$A + Q_{吸} - Q_{放} = \Delta E$			

$$C = \frac{dQ}{dT}$$

$$C_v = \frac{\frac{1}{2} \nu R dT}{dT} = \frac{1}{2} \nu R = \frac{R}{\gamma-1}$$

$$C_p = \frac{d(E+A)}{dT} = \frac{1}{2} \nu R + R$$

摩尔热容比 $\gamma = \frac{C_p}{C_v} = \frac{5}{3}$

单双原子V

多原子与T有关 $\left\{ \begin{array}{l} \text{振动} \\ \text{量子化} \end{array} \right.$

Δ 推导:

$$\left\{ \begin{array}{l} \nu C_v dT = 0 - p dV \\ p dV + \nu dT = \nu R dT \end{array} \right.$$

$$\Rightarrow C_v dV = - (C_v + R) p dV$$

$$\Rightarrow \frac{dV}{V} = - \gamma \frac{dV}{V}$$

$$\Rightarrow pV^\gamma = C$$

热机效率 (正循环)

$$\eta = \frac{|A|}{Q_{吸}} = \frac{Q_{吸} - Q_{放}}{Q_{吸}}$$

卡诺 $\eta = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$

致冷系数 (逆-)

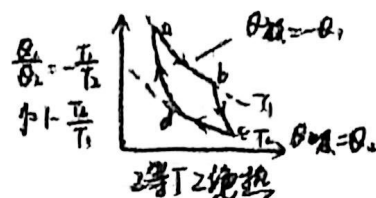
$$\epsilon = \frac{Q_{放}}{A} = \frac{Q_{放}}{Q_{吸} - Q_{放}}$$

$$= \frac{T_2}{T_1 - T_2}$$

奥托循环内燃机



卡诺循环理想热机



推导

$$\left\{ \begin{array}{l} Q_1 = \nu R T_1 \ln \frac{V_2}{V_1} \\ Q_2 = \nu R T_2 \ln \frac{V_4}{V_3} \\ T_1 V_1^{\gamma-1} = T_2 V_3^{\gamma-1} \\ T_1 V_2^{\gamma-1} = T_2 V_4^{\gamma-1} \end{array} \right. \Rightarrow \frac{V_2}{V_1} = \frac{V_4}{V_3} \quad \frac{Q_1}{Q_2} = - \frac{T_1}{T_2} \Rightarrow \frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0$$

$$\eta = 1 - \frac{|Q_2|}{Q_1} = 1 - \frac{T_2}{T_1}$$

- ① T_1, T_2
- ② $T_1 \uparrow T_2 \downarrow \eta \uparrow$
- ③ $\eta < 1$

玻尔兹曼熵 $S = k \ln W$ W 宏观态热力学概率

多方

$$p dV + V dp = \nu R dT$$

$$C_p = C_v + R$$

$$dE = dQ + dA$$

$$\nu C_v dT = \nu C_p dT - p dV$$

缩dT $\left[n = \frac{C_p - C_v}{C_p - C_v} \right] \frac{dp}{p} + n \frac{dV}{V} = 0$

ϵR

$$C = \frac{n-1}{n-1} C_v = \frac{(n-1) R}{(n-1)(\gamma-1)}$$

推导 $dQ = dA + dE$

$$\Rightarrow C dT = C_p dT + p dV$$

$pV^\gamma = 0$

$pV = R T \rightarrow$ 用T表p dV

热二 开 = 单热源热机/第二永动机, 而不...
或 = 低 $\rightarrow$ 高, 而不...



10. 气体动理论

平衡态 状态参量/状态函数/图

基本假设

① 分子动：大量 + 不停碰 + 无规则

② 统计(平衡态 p, p 处处同)：空间均匀 $n = \frac{N}{V}$ + 各向同概率 $\bar{v_x} = \bar{v_y} = \bar{v_z} = \frac{\sum N_i v_{ix}}{N}$

③ 理想气体 (p 低 T 高)：质点 + 完全弹性碰 + 相互作用为 0 $\rightarrow$ 范德瓦耳斯 $\left(\begin{matrix} \text{体积} \\ \text{相互作用} \end{matrix} \right)$

涨落、宏观瞬时间值 - 统计平均值

每个分子不可压缩体积为

$$\frac{1}{6}(\frac{4}{3}\pi d^3) \Rightarrow b = N_A \cdot 4V_0$$

一系列状态公式

$$dF = \sum_{i=1}^N m(v_{ix} ds) \cdot v_{ix} = \sum_i m_i v_{ix}^2 ds \quad p = \frac{dF}{ds} = n m \bar{v_x^2} = \frac{1}{3} n m \bar{v^2} = \frac{2}{3} n \bar{\epsilon_t}$$

$$\bar{\epsilon_t} = \frac{1}{2} m \bar{v^2} = \frac{3}{2} kT \quad \text{规定} \quad k = 1.38 \times 10^{-23} \text{ J/K} \quad \text{分子无规则热运动对系统质心的平均平动动能}$$

$$\bar{v^2} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} \quad R = N_A k = 8.31 \text{ J/(mol} \cdot \text{K)} \quad \text{气体摩尔常量} \quad \text{范德... 常量 } a, b \quad \text{不断压缩}$$

$$p = nkT \Rightarrow pV = \nu RT \Rightarrow \text{范德瓦耳斯} [p + a(\frac{\nu}{V})^2](V - \nu b) = \nu RT \quad \text{附加压强 } p_i = a(\frac{\nu}{V})^2 \quad V = \nu b$$

$$\text{能量均分原理: 每个自由度 } \frac{1}{2} kT \quad \text{分子平均能量 } \bar{\epsilon} = \frac{1}{2} kT \quad \text{内能 } E = \nu N_A \bar{\epsilon} = \nu \frac{1}{2} RT$$

分析自由度 $\begin{matrix} 6 \text{ 个力学} \\ \text{单分子} \end{matrix}$

弹性双原子分子 = 3 个平动 + 2 个转动 + 1 个振动 $\begin{matrix} E_p \\ E_k \end{matrix}$ Δ 振动的 1 个力学... = 2 个热力学...

微观量计算

$$n = \frac{N}{V} \quad \nu = \frac{m}{M} \quad \text{分子间平均距离 } N^{1/3} = V^{1/3}$$

单原子 $i=3$ 刚性双原子 $i=5$
双原子 $i=6$ 刚性多原子 $i=6$

麦克斯韦速率分布律

$$N \sim \nu \quad f(v) = \frac{dN}{N dv} \quad \int_0^\infty f(v) dv = 1$$

$$\nu \sim \nu dv \quad \frac{dN}{N} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \cdot e^{-\frac{mv^2}{2kT}} \cdot v^2 dv$$

$$\begin{cases} V_p \quad \frac{df(v)}{dv} \Big|_{V_p} = 0 \\ \bar{v} = \frac{\int v dN}{N} = \int_0^\infty v f(v) dv \\ \sqrt{\bar{v^2}} = \sqrt{\int_0^\infty v^2 f(v) dv} \end{cases}$$

理想

$$V_p = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}}$$

$$\frac{k}{m} = \frac{R}{M}$$

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}$$

$$\sqrt{\bar{v^2}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$



玻尔兹曼分布律

$$\text{等温变压公式} \quad dp = -\rho g dh \quad p = \frac{\rho M}{RT} \Rightarrow p = p_0 e^{-\frac{\rho g h}{RT}} = n kT$$

$$n = \frac{p_0}{kT} e^{-\frac{\rho g h}{RT}} = n_0 e^{-\frac{\rho g h}{RT}} = n_0 e^{-\frac{\epsilon_p}{kT}}$$

减小
↑ 快
↑ 慢

$$\text{分子} \begin{matrix} E_k \\ E_p \end{matrix} \quad \nu \quad \text{位置} \quad dN_{\nu, r} = n_0 \left(\frac{m}{2\pi kT} \right)^{3/2} \cdot e^{-\frac{\epsilon}{kT}} \cdot d\nu_x d\nu_y d\nu_z dx dy dz$$

$e^{-\frac{\epsilon}{kT}}$ 玻尔兹曼因子

$$N \sim \epsilon \quad \text{能量量子化} \quad N_i = A e^{-\frac{\epsilon_i}{kT}}$$

微观现象

$$\text{平均碰撞频率} \quad \bar{\nu} = \pi d^2 n \bar{v} \quad \text{其余动} \quad \sqrt{2} \pi d^2 n \bar{v}$$

$$\text{平均自由程} \quad \lambda = \frac{\bar{v}}{\bar{\nu}} = \frac{1}{\sqrt{2} \pi d^2 n} \quad \text{理想} \quad \frac{kT}{\pi d^2 p} \quad \text{温度} \uparrow \lambda \downarrow$$

如容器线度

(非平衡态)

迁移现象

$$\text{热传导} \quad dq = -k \frac{dT}{dx} ds dt$$

$$k = \frac{1}{6} k_B n \bar{v} \lambda$$

$$\text{黏滞} \quad dp = -\eta \frac{dv}{dx} ds dt$$

$$\eta = \frac{1}{3} n m \bar{v} \lambda = \frac{1}{3} \rho \bar{v} \lambda$$

$$\text{扩散} \quad dm = -D \frac{dc}{dx} ds dt \quad D = \frac{1}{3} \bar{v} \lambda$$



考点一览

易错偏

质点

运动学

① $r \rightleftharpoons v, a$

② $av = \frac{dv}{dt} \quad a_n = \frac{v^2}{r} \quad a = \sqrt{a_t^2 + a_n^2}$

③ 相对运动

$a = \frac{dv}{dt} = v \frac{dv}{dx}$

$\frac{1}{dt} = \frac{v}{dx}$

先求任意 t 的 v 公式

质点

动力学

① 牛 $F_{(x)} = m \frac{dv}{dt} \quad F_{(v)} = m \frac{dv}{dt} \quad F_{(m)} = m \frac{dv}{dt} \Rightarrow \int \frac{F_{(x)}}{m} dx = \int v dv$

② 惯性系 $\vec{F} + m\vec{a} = m\vec{a}$

③ 质心 $\vec{F}_{外} = m\vec{a}_c \quad r_c = \frac{\int \vec{r} dm}{\int dm}$

④ 密舍尔斯基 $m \frac{d\vec{v}}{dt} = \vec{F} + (\vec{v} - \vec{v}') \frac{dm}{dt}$

⑤ 动量冲量 $I_x = \int_{t_1}^{t_2} F_x dt = mv_{2x} - mv_{1x}$ 方向

⑥ 变力功 $A = \int \vec{F} \cdot d\vec{r}$

⑦ 机械能 $E_p = \int_p \vec{F} \cdot d\vec{r}$

⑧ 角动量 $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \quad \vec{M}_{外} = \frac{d\vec{L}}{dt} \quad \vec{M} = \vec{r} \times \vec{F}$

证保守力

$\oint \vec{F} \cdot d\vec{r} = 0$

$A = \int_{R_1}^{R_2} -\frac{Gmm}{r^2} dr = Gmm(\frac{1}{R_2} - \frac{1}{R_1})$

引力 $E_p = \int_r^\infty -\frac{Gmm}{r^2} dr = -\frac{Gmm}{r}$

$\vec{F} = -\frac{\partial E_p}{\partial x} \vec{i} - \frac{\partial E_p}{\partial y} \vec{j} - \frac{\partial E_p}{\partial z} \vec{k}$

$A_{外} + A_{保内} = (E_{k2} + E_{p2}) - (E_{k1} + E_{p1})$

$\Sigma A = 0 \Rightarrow \Delta E_k = -\Delta E_p$

刚体

力学

① 转动惯量 $J = \sum m_i r_i^2 = \int r^2 dm \quad J = J_c + mh^2$

② 转动定律 $M = J\beta + F = ma$

③ 能 $E_k = \frac{1}{2} J \omega^2 = \frac{1}{2} J \omega^2 + \frac{1}{2} mv_c^2 \quad E_p = mgh_c$

④ 角动量 $\vec{L} = J\vec{\omega} \quad \vec{M}_{外} = 0 \quad \vec{L} = \text{const}$

⑤ 平面运动 $\begin{cases} \Sigma \vec{F}_{外} = m\vec{a}_c \\ \Sigma M_c = J_c \beta \end{cases}$ 纯滚动 $\begin{cases} x_c = R\theta \\ v_c = R\omega \\ a_c = R\beta \end{cases}$ 方向判断

P 点无滑动/线速度为 0/静不做功

⑥ 进动 旋进 $\Omega = \frac{M}{L \sin \varphi} = \frac{mgr_c}{J\omega}$

⑦ 流体力学 $\begin{cases} \text{理想流体} = \text{不可压缩, 无黏滞性} \\ \text{定常流动} = v = \text{const} \end{cases}$

连续性方程 $S_1 v_1 = S_2 v_2 = C$
伯努利方程 $P + \frac{1}{2} \rho v^2 + \rho gh = C$ h 变化不大 $P + \frac{1}{2} \rho v^2 = C$

相对论 ① 运动学

洛伦兹

爱因斯坦

$\begin{cases} x' = \frac{x - vt}{\sqrt{1 - (v/c)^2}} \\ y' = y \\ z' = z \\ t' = \frac{t - v/c^2 x}{\sqrt{1 - (v/c)^2}} \end{cases}$

$\begin{cases} u_x' = \frac{u_x - v}{1 - u_x v/c^2} \\ u_y' = \frac{u_y \sqrt{1 - \beta^2}}{1 - u_x v/c^2} \\ u_z' = \frac{u_z \sqrt{1 - \beta^2}}{1 - u_x v/c^2} \end{cases}$

长度收缩

$l_0 = x_1 - x_2 \quad l' = (x_1 - x_2) \sqrt{1 - (v/c)^2}$

(同时)

$l' = l_0 / \gamma$

时间膨胀

$\Delta t_0 = t_2 - t_1 \quad \Delta t' = (t_2 - t_1) \sqrt{1 - (v/c)^2}$

(同地)

$\Delta t' = \Delta t_0 / \gamma$

不同时/地

$x_2 - x_1 = \frac{\Delta x' + u \Delta t'}{\sqrt{1 - (v/c)^2}} = \gamma (\Delta x' + u \Delta t')$

$\gamma = \frac{1}{\sqrt{1 - (v/c)^2}}$

$t_2 - t_1 = \frac{\Delta t' + u \Delta x'/c^2}{\sqrt{1 - (v/c)^2}} = \gamma (\Delta t' + u \Delta x'/c^2)$

$x_2' - x_1' = \gamma (\Delta x - u \Delta t)$

$t_2' - t_1' = \gamma (\Delta t - u \Delta x/c^2)$

② 动力学

$\begin{cases} \text{质速} m = \frac{m_0}{\sqrt{1 - (v/c)^2}} \\ \text{质能} E = mc^2 = m_0 c^2 + E_k \quad A = m_2 c^2 - m_1 c^2 \\ \text{能动} E^2 = p^2 c^2 + m_0^2 c^4 \end{cases}$



机械 ① 简谐表达式 $x = A \cos(\omega t + \varphi)$ $\omega = \sqrt{\frac{k}{m}}$ $A = \sqrt{x_0^2 + (\frac{v_0}{\omega})^2}$ $\tan \varphi = -\frac{v_0}{\omega x_0}$ $T = 2\pi \sqrt{\frac{m}{k}}$

振动

旋转矢量图：质点从一位置到另一位置最短时间

$$\begin{cases} v < 0 & 0 < \varphi < \pi \\ v > 0 & \pi < \varphi < 2\pi \\ v = 0 & x = A \ \varphi = 0 / x = -A \ \varphi = \pi \end{cases}$$

② 判定简谐

平衡 $\rightarrow$ 任意 $F = -kx$ $\frac{d^2x}{dt^2} = -\omega^2 x$

$\frac{1}{2} a \dot{x}^2 + \frac{1}{2} b (\frac{dx}{dt})^2 = \text{const}$ $F = -\frac{dE_p}{dx} = F_0 + \frac{dE_p}{dx}|_{x=0} \cdot x + \dots$ $k = \frac{d^2E_p}{dx^2}|_{x=0}$

模型：单摆 $\omega = \sqrt{\frac{g}{l}}$ 复摆 $\omega = \sqrt{\frac{mgJ}{J}}$ 扭摆 $\omega = \sqrt{\frac{k}{J}}$ 振子 $\omega = \sqrt{\frac{k}{m}}$

③ 振动合成

$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$

$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$

$\Delta \varphi = \begin{cases} \pm 2k\pi \\ \pm (2k+1)\pi \end{cases}$

$y = A [\cos(\omega t + \frac{2\pi}{\lambda} x + \varphi_1) + \cos(\omega t + \frac{2\pi}{\lambda} x + \varphi_2)]$

机械波 ① 波动表达式

$y = A \cos[\omega(t \mp \frac{x}{v}) + \varphi]$ 代已知点/原点 $\pm \frac{x}{v} / \Delta \varphi$

② 波的干涉

$\Delta \varphi = \varphi_2 - \varphi_1 - 2\pi \frac{r_2 - r_1}{\lambda}$ $\begin{cases} A_1 + A_2 \\ A_1 - A_2 \end{cases}$

③ 驻波

反射点入波 $\rightarrow$ 反波 相邻波节 $\Delta x = \frac{\lambda}{2}$

④ 多普勒

$\nu_r = \frac{u \pm v_r}{u \pm v_s} \nu_s$ 移动反射面 (≤ 1 次)

山谷回声

$\lambda' = (u - v) T_s$

$\nu_r = \frac{u + v}{\lambda'} = \frac{u + v}{u - v} \nu_s$

气体

① $pV = \frac{m}{M} RT$ $p = nkT = \frac{2}{3} n \bar{\epsilon}_t$ $\bar{\epsilon}_t = \frac{1}{2} m \bar{v}^2 = \frac{3}{2} kT$

分子动

$E = \frac{1}{2} kT = \frac{1}{2} pV$

理论

② $f(v) = \frac{dN}{N dv}$ $\int_0^\infty f(v) dv = 1$ $\bar{v} = \frac{\int_0^\infty v f(v) dv}{\int_0^\infty f(v) dv}$

③ $\bar{v} = \sqrt{2} \pi n d^2 \bar{v}$ $\bar{\lambda} = \frac{\bar{v}}{\bar{v}} = \frac{1}{\sqrt{2} \pi n d^2} = \frac{kT}{\sqrt{2} \pi n d^2 p}$

	平	转
单	3	0
双	3	2
多	3	3

热力学

① $Q = \nu C_V \Delta T$ $C_V = \frac{5}{2} R$ $C_p = C_V + R = (\frac{7}{2} + R)$

基础

$\Delta E = \nu C_V \Delta T = \frac{1}{2} (p_2 V_2 - p_1 V_1)$

$A = \int_{V_1}^{V_2} p dv$

$\Delta E = \frac{p_2}{2} V_2 - \frac{p_1}{2} V_1$

② 循环

正 $\eta = \frac{A}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$
逆 $\omega = \frac{Q_2}{A} = \frac{Q_2}{Q_1 - Q_2}$

卡诺 $= \frac{T_1 - T_2}{T_1}$
 $= \frac{T_2}{T_1 - T_2}$

T 不变 Q 不变 $\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0$

③ $\oint \frac{dQ}{T} = 0$ 可逆循环

$\Delta S = S_b - S_a = \int_a^b \frac{dQ}{T}$ 可逆过程

等 V $\Delta S = \nu C_V \ln \frac{T_2}{T_1}$ 等 T $\Delta S = \nu R \ln \frac{V_2}{V_1}$ 或 $\frac{p_1}{p_2}$
等 p $\Delta S = \nu C_p \ln \frac{T_2}{T_1}$ 绝热 $\Delta S = 0$

相变 $\Delta S = \frac{Q}{T}$

$Q = cm \Delta T$

静电场

叠加+高斯 $\phi = \oint \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum q_i$

球体 $E = \begin{cases} \frac{Qr}{4\pi\epsilon_0 R^3} & r < R \\ \frac{Q}{4\pi\epsilon_0 r^2} & r > R \end{cases}$ 壳 0

导线 $E = \frac{\lambda}{2\pi\epsilon_0 r}$ 半 $E = \frac{\lambda}{4\pi\epsilon_0 r}$

圆环 $E = \frac{\lambda}{2\pi\epsilon_0 R}$

平面 $E = \frac{\sigma}{2\epsilon_0}$ $E = \begin{cases} 0 & \text{外} \\ \frac{\sigma}{\epsilon_0} & \text{内} \end{cases}$

圆柱面 $E = \begin{cases} 0 & r < R \\ \frac{\sigma}{\epsilon_0} & r > R \end{cases}$ 半 $E_x = \frac{\sigma}{\epsilon_0}$ 轴

